Interactive Learning Assistant with Customizable 3D Avatars

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1. Introduction

The Interactive Learning Assistant project aims to enhance engagement and motivation in online education through the use of customizable 3D avatars. Built using Three.js and React, it provides students with an immersive learning experience by integrating emotional interactions and feedback. The tool addresses the challenges of sustaining student focus and motivation in remote learning environments, offering features such as text-to-speech reading, guided study prompts, and avatar expressions that respond to user performance.

2. Planning

2.1 Timing

Project Setup and Research

- Define project purpose and scope.
- Conduct research on 3D modeling, interactive design principles, and educational engagement techniques.
- Choose suitable development tools and resources, including Three.js and React.
- Set up the development environment, including necessary libraries.

Prototype

- Develop the core avatar interaction module.
- Implement real-time text-to-speech and emotional feedback systems.
- Create the customizable avatar interface.

Testing and Optimization

- Analyze user feedback and optimize user interaction.
- Develop documentation, including user manuals, technical specifications, and code comments.
- Finalize the project and upload to a web repository.

2.2 Scope

1- Interactive Avatars: Developing a customizable 3D avatar system that enhances user engagement.

2- Emotional Feedback: Implementing a real-time feedback mechanism using avatars to improve learning outcomes.

3- Integration with LMS: Ensuring compatibility with existing learning management systems or as a standalone tool.

4- Accessibility: Incorporating text-to-speech features for enhanced accessibility.

2.3 Technology Used

• Three.js: A JavaScript library used for 3D graphics.

• React: A JavaScript library for building user interfaces.

• Text-to-Speech API: For real-time audio feedback.

• Development Environment: Visual Studio Code or similar IDEs for efficient development and debugging.

3. Purpose of the Project

The primary purpose of the Interactive Learning Assistant project is to enhance the online learning experience by incorporating customizable 3D avatars that provide emotional and performance-based feedback. This tool aims to address the challenges of remote learning by making education more engaging, personalized, and accessible.